Practice Mock Exam

🎓 Artificial Intelligence (AI2002) - Comprehensive Mock Final

Total Time: 3 Hours | **Total Marks:** 50

Question 1: Intelligent Agents, Environments, & Search (10 Marks)

Part A (4 Marks): You are designing an AI agent for a Mars Rover. Its goal is to navigate a crater, drill for rock samples, and avoid falling into ravines. Dust storms can suddenly reduce visibility, and the rover's wheels sometimes slip on the loose Martian soil. It communicates with a separate orbiting satellite agent to map the terrain.

Identify the environment type for this rover across the following 4 dimensions and provide a one-line justification for each:

1. Fully Observable vs. Partially Observable
2. Deterministic vs. Stochastic
3. Static vs. Dynamic
4. Single-Agent vs. Multi-Agent

Part B (3 Marks): In the context of adversarial search (game theory), explain how the Alpha-Beta Pruning algorithm improves upon standard Minimax. Does Alpha-Beta pruning ever change the final decision the agent makes compared to standard Minimax? Justify your answer.

Part C (3 Marks): You are formulating a Constraint Satisfaction Problem (CSP) for a university timetable. You have 3 classes (C_1, C_2, C_3) and 2 time slots (T_1, T_2). C_1 and C_2 are taught by the same professor. C_2 and C_3 require the same specialized lab equipment.

Define the **Variables**, their **Domains**, and write out the explicit **Constraints** required for this CSP.

Question 2: Probabilistic Reasoning (10 Marks)

Part A: Bayesian Networks (5 Marks)

Consider a smart-home security system. A Power Outage (O) and a System Glitch (G) can both cause the House Alarm (A) to trigger. If the Alarm triggers, it can cause the neighbor to Call the Police (P).

- $P(O) = 0.1$
- $P(G) = 0.2$
- $P(A \mid O, G) = 0.95$
- $P(A \mid O, \neg G) = 0.80$
- $P(A \mid \neg O, G) = 0.85$
- $P(A \mid \neg O, \neg G) = 0.05$
- $P(P \mid A) = 0.90$
- $P(P \mid \neg A) = 0.10$

Calculate the probability that the Police are called (P), given that there is NO Power Outage ($\neg O$) and NO System Glitch ($\neg G$). *(Show your step-by-step formula expansion using the conditional independence rules of the network).*

Part B: Naïve Bayes (5 Marks)

An e-commerce website uses a Naïve Bayes classifier to predict if a user will "Buy" a product based on their browsing behavior.

Device	Time of Day	Cart Items	Buy (Target)
Mobile	Morning	Low	No
Desktop	Evening	High	Yes
Mobile	Evening	Low	No
Desktop	Morning	High	Yes
Mobile	Evening	High	Yes
Desktop	Morning	Low	No

Using the Naïve Bayes algorithm, calculate the probability scores to predict whether a new user will Buy or Not Buy given the following profile:

- Device: Mobile
- Time of Day: Morning
- Cart Items: High

Question 3: Supervised Learning - Regression & KNN (10 Marks)

Part A: Linear Regression (5 Marks)

A data center wants to predict its monthly cooling cost (Y, in hundreds of dollars) based on the average external temperature (X, in Celsius).

Temperature (X)	Cooling Cost (Y)
20	4
25	6
30	8
35	9

Calculate the linear regression line ($Y = mX + b$). Once you have the equation, predict the cooling cost if the external temperature reaches **40 degrees Celsius**.

(Hint: You will need to calculate $\sum X, \sum Y, \sum XY, \sum X^2$).

Part B: K-Nearest Neighbors (5 Marks)

A botanical lab is classifying a new species of poisonous plant based on Leaf Width (X_1) and Toxin Level (X_2).

Plant	Leaf Width (X1)	Toxin Level (X2)	Class
P1	2.0	4.0	Safe
P2	4.0	8.0	Toxic
P3	3.0	3.0	Safe
P4	6.0	9.0	Toxic

A newly discovered plant has a Leaf Width of **4.0** and a Toxin Level of **5.0**.

Calculate the Euclidean distance between the new plant and all 4 training samples. Using $K = 3$, what is the predicted class of the new plant?

Question 4: Decision Trees & Model Evaluation (10 Marks)

Part A: Decision Trees / ID3 (6 Marks)

A bank uses a decision tree to approve loans.

Income	Credit Score	Approve Loan
High	Good	Yes
Low	Bad	No
High	Bad	No
Low	Good	Yes

Income	Credit Score	Approve Loan
High	Good	Yes

1. Calculate the initial Entropy of the entire dataset for the target variable "Approve Loan".
2. Calculate the Information Gain for the attribute "Credit Score".

Part B: Confusion Matrix (4 Marks)

An AI model is deployed to detect fraudulent credit card transactions. Out of 1,000 total transactions evaluated:

- The model flagged 60 transactions as Fraud.
- Out of those 60, 45 were actually fraudulent, but 15 were legitimate.
- Out of the 940 transactions the model labeled as Safe, 10 were actually fraudulent.

Draw the Confusion Matrix (labeling TP, TN, FP, FN). Then, calculate the Precision, Recall, and overall Accuracy of the model.

Question 5: Artificial Neural Networks & Clustering (10 Marks)

Part A: Artificial Neural Networks (6 Marks)

You are designing a feedforward neural network to predict if a student will pass a test. The network has 2 input nodes (x_1, x_2), 1 hidden node (h_1), and 1 output node (O).

- **Inputs:** $x_1 = 0.5$, $x_2 = 1.0$
- **Weights (Input to Hidden):** $w_{11} = 0.8$, $w_{21} = -0.4$
- **Bias for Hidden Node:** $b_1 = 0.1$
- **Weight (Hidden to Output):** $w_{h1} = 0.6$
- **Bias for Output Node:** $b_2 = -0.2$

Perform a Forward Pass to calculate the final value of the Output node (O). Use the ReLU activation function for the hidden layer, and the Sigmoid activation function for the output layer. *(Assume $e^{0.1} \approx 1.10$ or similar if needed, or leave it in fraction form).*

Part B: K-Means Clustering (4 Marks)

You are given 4 data points on a 1D axis: $A = 2, B = 4, C = 10, D = 12$.

You are running K-Means clustering with $K = 2$. Your initial, randomly chosen cluster centroids are $C_1 = 2$ and $C_2 = 4$.

Perform one complete epoch/iteration of K-Means.

1. State which data points are assigned to C_1 and which to C_2.

2. Calculate the updated values for the centroids C_1 and C_2 after the assignments.